

PAPER #91 ■ MUGE Resonance: 14-Mode Gravity Framework

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Title: MUGE Resonance Gravity: 14-Mode Framework from aDPM Base Through Wormhole Metric

Author: Daniel T. Murphy **Framework:** MUGE Resonance, UQFF Star-Magic ([SSq] = 0.57, [SCm] ■ 0.99) **Date:** March 7, 2026 **Source Data:** validateuqffmuge.py, source4.cpp (14 Resonance functions), computeresonanceMUGE_SOURCE4 **Index Slot:** ■1.12 UQFF Master Calculators, Paper #91

Abstract

MUGE Resonance extends compressed gravity with 14 mode-specific corrections, beginning from the anomalous Doppler modulation (aDPM) base and including terahertz, vacuum diffusion, super-frequency, aether resonance, Ug4 intensity, quantum frequency, aether frequency, fluid frequency, oscillation, expansion frequency, Toroidal Resonance Zone, and wormhole metric contributions. The compute_resonance_MUGE_SOURCE4 function returns a complete resonance gravity value for any astrophysical system; validate_uqff_muge.py confirms all 14 terms are finite across 5 systems.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^4$ day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. The 14-Mode Resonance Decomposition

MUGE Resonance gravity:

$$g_{\rm MUGE}^{\rm Res}(r, \vec{\omega}) = g_{\rm aDPM}(r) + \sum_{k=1}^{13} \delta_k^{\rm Res}(r, \vec{\omega})$$

Mode k	Symbol	Frequency/Origin	Physical Effect
Base	aDPM	Anomalous Doppler	Doppler-modified gravity
1	aTHz	THz frequency	Terahertz coupling
2	Avac_diff	Vacuum diffusion	Vacuum polarization diffusion
3	aSuperFreq	SuperFreq (SGR1745)	Magnetar super-frequency
4	aAetherRes	Aether resonance	Quantum vacuum resonance
5	Ug4_i	Ug4 intensity mode	BH vacuum concentration
6	aQuantumFreq	QuantumFreq	Quantum oscillation modes
7	aAetherFreq	AetherFreq	Aether field frequency
8	aFluidFreq	FluidFreq	Navier-Stokes fluid resonance
9	Osc_term	General oscillation	Composite oscillation
10	aExpFreq	ExpFreq	Hubble expansion frequency

Mode k	Symbol	Frequency/Origin	Physical Effect
11	fTRZ	Toroidal Resonance Zone	f_TRZ = 0.01
12	Wormhole	Wormhole metric	Einstein-Rosen bridge topology

2. aDPM Base Grace

The anomalous Doppler modulation base:

$$g_{\rm aDPM}(r, v) = \frac{GM}{r^2} \cdot \frac{1 - v/c}{1 + v/c}$$

For bound circular orbital: $v = (GM/r)^{1/2}$, giving:

$$g_{\rm aDPM}(r) = g_{\rm Newton}(r) \cdot \left(1 - 2\sqrt{R_S/r}\right)^{1/2}$$

? At $r = 10 R_S$: correction = -6.3% (sub-GR post-Newtonian)

3. 5-Frequency Resonance (Source27/28 Origin)

Modes 1,3,6,7,8 correspond to the 5-frequency SuperFreq resonance from source27/28:

Frequency	Symbol	Origin System
SuperFreq	aSuperFreq	SGR1745 Magnetar
QuantumFreq	aQuantumFreq	SgrA*
AetherFreq	aAetherFreq	Universal vacuum
FluidFreq	aFluidFreq	Accretion disk
ExpFreq	aExpFreq	Hubble expansion

Combined 5-frequency resonance amplitude:

$$A_{\rm 5freq} = \prod_k \left(1 + a_k \cos(\omega_k t)\right) \approx 1 + \sum_k a_k \cos(\omega_k t)$$

For small amplitudes $a_k \ll 1$.

4. TRZ Mode (fTRZ Factor)

The Toroidal Resonance Zone mode:

$$\Delta_{\rm TRZ}(r) = f_{\rm TRZ} \cdot g_{\rm aDPM}(r) = 0.01 \times g_{\rm aDPM}(r)$$

This is the same $f_{\rm TRZ} = 0.01$ that modifies Hawking temperature (Paper #81) ■ a universal UQFF factor. At the horizon scale, $d_{\rm TRZ} = 0.01$ ■ ?g provides a 1% enhancement observable in precision pulsar timing.

5. Wormhole Metric Contribution

The wormhole mode uses Ellis drainhole metric:

$$\Delta_{\rm WH}(r) = g_{\rm aDPM}(r) \cdot \exp\left(-\frac{r^2}{L_{\rm WH}^2}\right)$$

Where $l_{WH} = \text{Planck-scale wormhole throat}$. For $r \gg l_{WH}$: $d_{WH} \approx 0$ (undetectable). For Planck-regime: $d_{WH} \approx g_{aDPM}$ (full wormhole topology correction).

6. Validation Results (validateuqffmuge.py)

All 14 resonance modes computed for all 5 systems:

System	$g_{total}^{\wedge}Res$ (m/s ²)	All modes finite	aDPM correction
Sgr A* (r_horizon)	238.4	?	-1.7%
M87* (r_horizon)	2261	?	-1.9%
Sun (surface)	273.8	?	-0.003%
NeutronStar (surface)	1.63×10^{12}	?	-5.1%
Magnetar (surface)	1.75×10^{12}	?	-5.2%

7. Comparison: Compressed vs Resonance

Property	MUGE Compressed	MUGE Resonance
Terms	10 (static corrections)	14 (frequency-dependent)
Primary physics	Multi-scale corrections	Oscillation modes
Stable results	Always	When ω_k bounded
Dominant regime	Galaxy/cosmological	Near-compact object
TRZ included	No (Compressed uses d_{Ug4} only)	Yes (explicit fTRZ mode)
Wormhole	No	Yes (Planck-scale)

Summary

The MUGE Resonance 14-mode framework provides the most complete gravity description for compact object environments, combining the 5-frequency resonance from source27/28 with TRZ, aDPM Doppler correction, and Planck-scale wormhole topology. All 14 modes are finite for 5 astrophysical systems.

Source: validateuqffmuge.py | source4.cpp computeresonanceMUGE_SOURCE4 | 14 modes $\approx$ 5 systems all finite

See also: PAPER_090 | Part of the Star-Magic UQFF Whitepaper Series.

UQFF computed: MUGE buoyancy ratio $U_{bi}/F_U = [SSq] \omega^2 r^3 / GM = 5.7e-1 \times 5.0e-4 = 2.85e-4$; compressed MUGE baseline $g = 5.4e-7$ m/s² at r_{ISCO} .

Appendix: UQFF Production Framework Reference (v4.75+)

*Added by upgradeearlywhitepapers.py (v4.75). This appendix cross-references
the production physics constants and master equations to enable reproducibility
against the current codebase state.*

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60–0.61	Buoyancy coupling coefficient
k_{DPM}	1.5	Ug1 DPM-dipole coupling
k_{out}	1.2	Ug2 outer-bubble charge coupling
k_{str}	1.8	Ug3 string-rotation coupling
k_{vac}	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
$E_{\text{react}}(0)$	10^{-10} J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

$$F_U = U_{\{g1\}} + U_{\{g2\}} + U_{\{g3\}} + U_{\{g4\}} + U_{\{bi\}} + U_m - \sum_{i=1}^4 \text{igl}[\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}}] \text{igr}$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	<code>compute_Ug1_SOURCE4 / compute_Ug1()</code>
Ug2	Outer-field bubble (charge-reactivity)	<code>compute_Ug2_SOURCE4 / compute_Ug2()</code>
Ug3	Magnetic string rotation	<code>compute_Ug3_SOURCE4 / compute_Ug3()</code>
Ug4	Vacuum concentration (star-BH)	<code>compute_Ug4_SOURCE4 / compute_Ug4()</code>
Ubi	Buoyancy force	<code>compute_Ubi_SOURCE4 / compute_Ubi()</code>
Um	Universal Magnetism (Heaviside-amplified)	<code>compute_Um_SOURCE4 / compute_Um()</code>
$-\sum \lambda_i \cdot U_i \cdot E_{\text{react}}$	4th dissipation term (PAPER_420)	<code>compute_FU_SOURCE4 / full pipeline</code>

4th dissipation term parameters (PAPER_420): $\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 U_m Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \cdot \text{igl}(1 + 10^{13}) \cdot \Theta(\rho_c - \rho) \cdot \text{igl}(1 + A_q \cos(\Delta\omega \cdot t)) \cdot \text{igr}$$

Symbol	Value	Description
ρ_c	10^1 kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\Delta\omega$	$2\pi/(434 \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	$U_g_{\text{sum}} + \text{Newtonian base}$	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	$\beta_i \times U_{\text{bi}}$	Expanding nebulae, stellar winds
Superconductive	$U_m \times (1 + 10^{13} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, and `CondensedPhysics2.py`.